

PAPER



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EPR spin trapping evidence of radical intermediates in the photo-reduction of bicarbonate/CO₂ in TiO₂ aqueous suspensions†

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Using the EPR spin trapping technique, we prove that simultaneous reactions take place in illuminated suspensions of TiO₂ in aqueous carbonate solutions (pH ≈ 7). The adsorbed HCO₃[−] is reduced to formate as directly made evident by the detection of formate radicals ([•]CO₂[−]). In addition, the amount of OH[•] radicals from the photo-oxidation of water shows a linear dependence on the concentration of bicarbonate, indicating that electron scavenging by HCO₃[−] increases the lifetime of holes. In a weakly alkaline medium, photo-oxidation of HCO₃[−]/CO₃^{2−} to [•]CO₃[−] interferes with the oxidation of water. A comparative analysis of different TiO₂ samples shows that formation of [•]CO₂[−] is influenced by factors related to the nature of the surface, once expected surface area effects are accounted for. Modification of the TiO₂ surface with noble metal nanoparticles does not have unequivocal benefits: the overall activity improves with Pd and Rh but not with Ru, which favours HCO₃[−] photo-oxidation even at pH = 7. In general, identification of radical intermediates of oxidation and reduction reactions can provide useful mechanistic information that may be used in the development of photocatalytic systems for the reduction of CO₂ also stored in the form of carbonates.

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Introduction

Carbon dioxide is the most abundant form of oxidized carbon occurring in nature. Its conversion to useful chemicals has been the aim of intense research over several decades and is presently experiencing an important surge of interest as concern is growing over the threat of greenhouse effects.^{1–3} Unfortunately, it is a prominent result of both experimental^{4–7} and theoretical⁸ studies that the reductive conversion of CO₂ is a difficult, challenging process. It is currently understood that [•]CO₂[−] is the key surface intermediate in both electro-catalytic and photocatalytic reduction on semiconductors



which can subsequently undergo multielectron reduction to formate, CO and further to methanol and methane.

The redox potential of reaction (1) is −1.9 V vs. NHE, and understandably electrochemical reduction is advantageous due to the possibility to span a large interval of potentials. Results of electrochemical studies focusing on application have already appeared in the literature^{9,10} although the competing evolution of hydrogen can be a limitation still requiring efforts towards an optimized choice of the metal electrode.⁶ Conversely, the redox potential is far too negative compared to the energy of photo-generated electrons on most semiconductors⁵ in aqueous media; in consequence, the occurrence of reaction (1) appears thermodynamically impossible in the photocatalytic approach.

Photocatalysis, on the other hand, has the important advantage of sustainability in that light is used as a continuous source of renewable energy for reactions to occur on stable and environmentally benign semiconductors such as TiO₂. It stands to reason, then, that research in this field continues to be very active,^{11–13} with particular attention being paid to activation of the rather inert CO₂ molecule. Indeed this is the most important step in any reaction of CO₂ basically involving the transformation of its linear structure into a more reactive bent form, i.e. in a state in which the LUMO is lowered enough to facilitate electron transfer.⁷ Conditions become favorable when CO₂ is coordinated to active centers of homogenous phase molecular catalysts or when it is adsorbed on surfaces.¹⁴ In this regard, earlier^{13,15} and recent^{16–19} studies have under-

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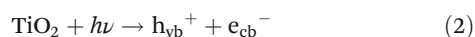
lined the advantages of CO₂ conversion to carbonate/bicarbonate which, upon adsorption, may undergo an easier reduction. In an interesting recent paper it has been shown that in CO₂-saturated bicarbonate solutions (pH 6.8), bicarbonate undergoes electrochemical reduction first to produce formate and, at higher applied potentials, CO₂ is reduced to CO.¹⁹

Reduction of CO₂ is a multielectron process that leads to formation of various products *via* different pathways that, among other factors, depend on the characteristics of the catalyst employed.¹¹ Basic questions that have been posed concern the dynamics of electron transfer and the identification of the active surface sites.⁷ Then, the analysis of reaction mechanisms has a fundamental importance since understanding of the basic reaction steps is a critical aspect in the design of selective and efficient systems. In particular one needs information about the initial steps of CO₂ reductive activation, which has been given less attention in comparison, for example, with processes such as the oxidation of water or alcohols. In this context, a recent electron paramagnetic resonance (EPR) study by Dimitrijevic *et al.*²⁰ is to be mentioned as an important contribution. The EPR technique offers indeed a potent tool for the detection of short-lived radical species.^{21–24}

These considerations are the motivation that led us to undertake the investigations described in the present paper. Specifically, we used the EPR spin-trapping technique to analyse directly the formation and the role of radical intermediates in illuminated deaerated TiO₂ aqueous suspensions containing carbonates, under photocatalytic conditions in which water oxidation and reduction processes are required to occur simultaneously.

Results and discussion

Photocatalysis on semiconductors, such as titanium dioxide, is based on the photo-generation of charge: positive holes in the valence band (h_{vb}^+) of the oxide and electrons in its conduction band (e_{cb}^-) according to eqn (2).



In order to achieve an efficient photocatalytic process, holes and electrons have to be quickly transferred to donor and acceptor species. Since oxidation and reduction processes occur on the same photoexcited particle, under steady state conditions the two rates must be equal on the basis of a charge balance requirement. In principle one can, for example, gain information about the reduction reaction by following the time course of the oxidation process, which indeed has been verified in some instances.²⁵ In the case of transient measurements, however, it might be an optimistic hypothesis since electrons or holes can accumulate during short time experiments instead of being transferred to acceptor or donor species, respectively. In the following, we will explore factors that influence the reduction of bicarbonate and the oxidation of water, which are closely connected processes taking place on different sites of the same photocatalyst particle.

Effect of NaHCO₃ on the formation of OH[•] radicals

In our deaerated aqueous NaHCO₃/CO₂ system, water oxidation by holes occurs with formation of OH[•] radicals (eqn (3)) and since direct reduction of bicarbonate is emphasized in several literature reports,^{13–20} one can assume in the first approach that the adsorbed HCO₃[–] is the electron scavenger (eqn (4)). Monitoring of hydroxyl radicals by the EPR spin trapping technique gives at least an indirect indication about reduction of carbonates.



The reported thermodynamic potential of the process described using eqn (4) is –0.52 V (ref. 13) and the mechanism seemingly involves concerted proton and electron transfer.²⁰ Regarding reaction (3), it is noteworthy that under short time illumination, the species trapping holes may be actually surface OH groups.

In our study we use α -phenyl *N*-tert-butyl nitron (pbn) as a spin trap. Its reaction with OH[•] radicals leads to a paramagnetic adduct [pbn–OH][•] that can be identified by both the hyperfine structure and the coupling constant values.²⁶ In the literature, 5,5-dimethyl pyrroline *N*-oxide (DMPO) is often used as a spin trap because the signal of the [DMPO–OH][•] adduct has peculiar characteristics that facilitate the identification of the trapped radical.^{23,24,26} However, we decided that DMPO is not reliable in our study because its adducts with ‘CO₂^{•–}’ and ‘CO₃^{•–}’ radical intermediates are not stable; after a fast rearrangement they actually evolve to [DMPO–OH][•], giving misleading information about the mechanism of radical formation in the photocatalytic process.²⁷

When deaerated aqueous suspensions (pH $\approx$ 7) of TiO₂-P25 containing pbn are illuminated, one observes EPR spectra consisting of a triplet of doublets (Fig. 1, curve 1) with coupling constants $a_{\text{N}} = 15.4$ G and $a_{\text{H}} = 2.8$ G. In agreement with the literature,²⁶ this signal is attributed to the paramagnetic adduct [pbn–OH][•], confirming the formation of OH[•] radicals (eqn (3)). An analogous experiment was carried out starting from deaerated suspensions containing NaHCO₃ (0.01 M, initial pH was 8.8, brought to pH 7.2–7.5 by bubbling of CO₂). The spectra recorded upon illumination reveal again the [pbn–OH][•] adduct (Fig. 1, curve 2) and, notably, signal intensities are greater when bicarbonate is present, under the same pH conditions. Although the trapping process is not quantitative, the amount of trapped radicals can be considered constant if an excess of spin trap is used.²⁸ On this basis, one can compare spectra obtained under identical experimental conditions and safely assume a proportionality between the signal intensity and the concentration of photo-generated radicals.

In addition, another signal is observed in Fig. 1 (marked *) consisting of a quartet with $a_{\text{N}} = a_{\text{H}} = 14.2$ G. In accordance with the literature,²⁹ this quartet is ascribed to trapped ‘CO₃^{•–}

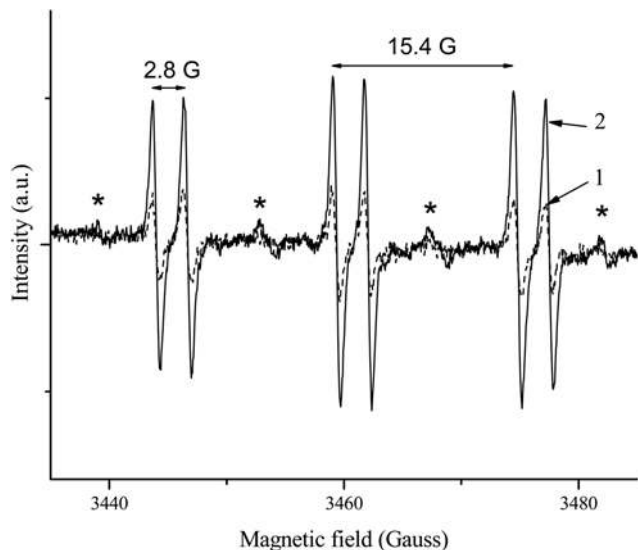


Fig. 1 EPR spectra obtained upon irradiation ($\lambda > 360$ nm) of deaerated aqueous suspensions of TiO₂-P25 containing pbn (0.1 M) at pH 7.2–7.5 in the absence (curve 1) and in the presence of NaHCO₃ (0.01 M)/CO₂ (curve 2).

radicals, which can be formed from reaction between bicarbonate ions and OH[•] radicals (eqn (5)):

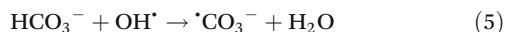


Photo-oxidation of carbonate/bicarbonate to [•]CO₃[−] radicals has been observed before²⁰ in the EPR spectra, but at 4.5 K.

Reduction of HCO₃[−]/CO₂ should be favored by a pH increase since the flat band potential becomes more negative according to

$$E_{\text{fb}} = E_{\text{fb}}(\text{pH} = 0) - 0.06 \text{ pH} \quad (6)$$

However, as pH is increased, photo-oxidation of HCO₃[−]/CO₃^{2−} (eqn (5)) can compete with its reduction (eqn (4)) as demonstrated by Sayama and Arakawa³⁰ who have also emphasized the role of [•]CO₃[−] radical intermediates in promoting O₂ evolution on TiO₂ at pH > 10 and described a reaction mechanism of [•]CO₃[−] formation from carbonate oxidation in an alkaline medium.

Our results showed that under our experimental conditions (pH < 8), the intensity of the signal due to the [pbn-CO₃[−]][•] adduct is at least 5-fold less than that at pH = 10 for the same illumination time (2 min). Importantly, no intensity increase with irradiation time was observed in neutral solutions, indicating that the reaction between HCO₃[−] and OH[•] radicals (eqn (5)) is here a minor pathway that does not proceed to any appreciable degree and does not interfere with trapping of other radicals. We also point out that the occurrence of reaction (5) to any significant extent would lead to an important decrease of the [pbn-OH][•] signal intensity that was not observed at pH < 8. In contrast, Fig. 2 shows that the [pbn-OH][•] intensity increases linearly as the NaHCO₃ concentration is increased.

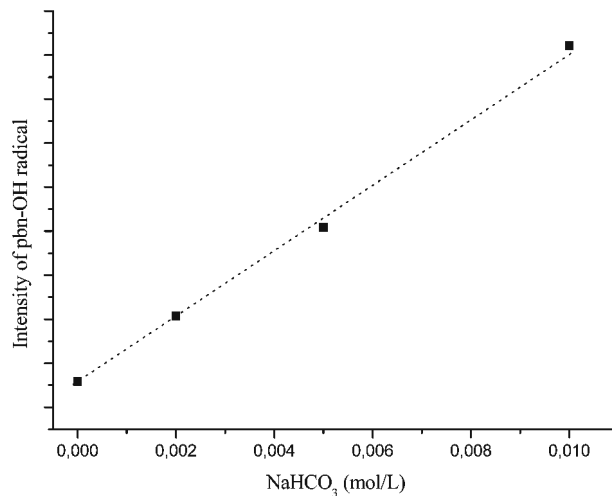
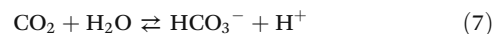


Fig. 2 Intensity of the [pbn-OH][•] adduct as a function of NaHCO₃ concentration obtained upon irradiation ($\lambda > 360$ nm) of TiO₂-P25 deaerated aqueous suspensions containing pbn (0.1 M). pH is maintained in the range 6.8–7.5 by bubbling CO₂.

Since the pH was kept at 7–7.5 in these experiments, HCO₃[−] constitutes more than 80% of all the inorganic carbon forms in the solution (CO₂/HCO₃[−]/CO₃^{2−}). The ratio between the species in the equilibrium



will remain constant in the dark while, under illumination, it might shift to the right side if HCO₃[−] is photo-reduced. In turn, consumption of electrons in this process will increase the lifetime of holes and favor the oxidation of water involving OH radical intermediates, as shown in Fig. 2.

We have performed additional experiments in which either the concentration of NaHCO₃ or pH was changed. The intensities of the [pbn-OH][•] adduct that is formed upon illumination of TiO₂-P25 in these media are reported in Fig. 3. Comparison of entries 1–3 confirms that an increase in the concentration of NaHCO₃ results in a corresponding increase of the [pbn-OH][•] intensity. Sample 4 refers to a system obtained by bubbling CO₂ through a bicarbonate solution until a final pH of 5.5. At this pH value about 70% of all the carbon forms is CO₂/H₂CO₃ and one notes a substantial increase of OH[•] radicals as revealed by the greater intensity of the [pbn-OH][•] adduct.

To a first analysis, this result would indicate that carbon dioxide is a very good electron acceptor, which would decrease recombination and in consequence enhance water oxidation by holes. However, we noted that the recorded [pbn-OH][•] intensity for illuminated TiO₂ aqueous suspensions saturated with CO₂ until pH 4.2 (entry 5) is comparable to that obtained in deaerated water (entry 1) and lower than that obtained at a pH of 5.5 (entry 4). At pH ≈ 4 there are no bicarbonate anions in equilibrium with carbon dioxide and this leads one to conclude that electron capture by dissolved CO₂ is inefficient, in accordance with the poor interaction of the unactivated mole-

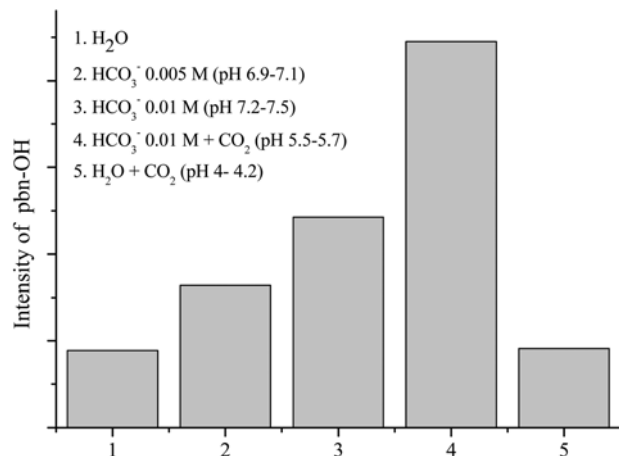


Fig. 3 Comparison of the intensities of the [pbn-OH] $\cdot$ adduct obtained upon irradiation of TiO₂-P25 deaerated aqueous suspensions containing pbn (0.1 M) with different concentrations of NaHCO₃. Columns 4 and 5 are relative to samples deaerated with CO₂.

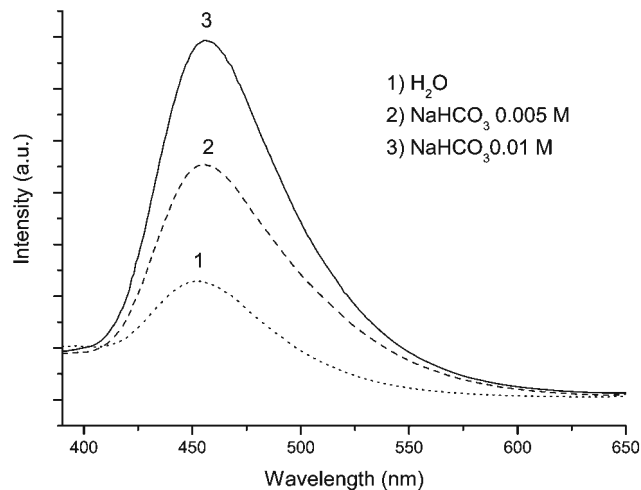


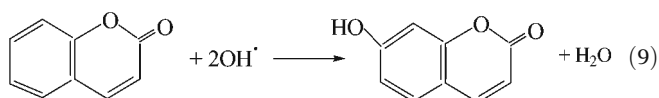
Fig. 4 Emission intensity of 7-HC measured at 455 nm obtained on illumination (15 min) of deaerated aqueous suspensions of TiO₂-P25 containing coumarin (1×10^{-4} M).

cule with TiO₂. The possible explanation for the higher amount of OH $\cdot$ radicals measured at pH = 5.5 (sample 4) is that this pH is lower than the point of zero charge (PZC $\sim$ 7 for P-25) and the TiO₂ surface charge is positive. This condition is expected to enhance adsorption of the bicarbonate anion, as shown in eqn (8),



and, in fact, adsorption is evident in a comparison of cyclic voltammetry curves (ESI Fig. 1†) recorded in NaHCO₃ at pH 7 and 5.5. In the latter case one notes new cathodic and anodic peaks that reflect significant changes in the band-gap states.

In order to buttress the conclusions of EPR results, we used the hydroxylation of coumarin as an independent method to verify the effect of bicarbonate on the formation of OH radicals.^{31,32}



Formation of 7-hydroxycoumarin (7-HC) (eqn (9)) can be followed by emission spectroscopy as it gives an intense luminescence upon UV light excitation. The results displayed in Fig. 4 clearly show that the intensity emission spectra depend on NaHCO₃ (curves 1–3) and, moreover, measurements at different concentrations showed that the dependence is linear, confirming the EPR data of Fig. 2. The emission intensity of 7-HC formed at pH $\approx$ 4 is significantly lower than that at pH 5.5 (ESI Fig. 2†) and, in agreement with EPR results (see above, entry 5 of Fig. 3), OH $\cdot$ radical formation is a comparatively less efficient process when CO₂/H₂CO₃ are the dissolved species.

In addition, as pH increases from 7 to 10 the fluorescence intensity increases in the absence of bicarbonate and decreases in its presence (ESI Fig. 3†). The lower OH $\cdot$ radical

formation caused by the carbonate is in accordance with reaction (5).

Evidence of formate production through detection of $\cdot\text{CO}_2^-$ radicals

The results reported above seemingly suggest that bicarbonate anions can act as the electron acceptors thus improving both charge separation and the efficiency of water photo-oxidation. This conclusion is consistent with experimental facts but it is based on indirect evidence of the photo-reduction of HCO₃ $^-$. In the following we will give proof of the occurrence of bicarbonate reduction to formate by showing the formation of $\cdot\text{CO}_2^-$ radicals that can be directly revealed by EPR spin trapping. Interestingly though, these radicals were not detected in previous work in which DMPO was used as the spin trap.²⁰

Fig. 5 displays the effect of the illumination time on the EPR spectra for a TiO₂-P25 suspension in a NaHCO₃ solution. In the short time scale (<60 s, curve 1) the only observed spectrum corresponds to the [pbn-OH] $\cdot$ adduct (*cf.* Fig. 1) while a new, well defined triplet of doublets, with hyperfine coupling constants of $a_N = 15.5$ G and $a_H = 4.7$ G, appears on prolonging irradiation (3 min). These last parameters correspond to the trapping of formate radicals $\cdot\text{CO}_2^-$ by pbn²⁶ and, interestingly, its detection might imply reduction of bicarbonate (eqn (4)).

One possibility would be that formate is accumulated in a sufficient amount to give $\cdot\text{CO}_2^-$ by reaction with OH $\cdot$ radicals ($k = 3.5 \times 10^9 \text{ M}^{-1} \text{ s}^{-1}$) according to eqn (10):



But actually, since experiments refer to the initial phenomena, occurring at a short illumination time, a more plausible reaction mechanism of $\cdot\text{CO}_2^-$ generation from bicarbonate reduction is proposed in Scheme 1, involving simultaneous transfer of electrons and protons.

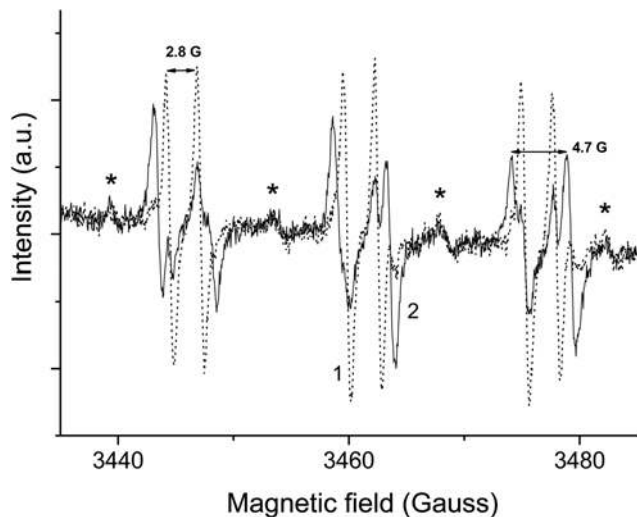


Fig. 5 EPR spectra obtained on illumination for 30 s (curve 1) and 3 min (curve 2) of deaerated TiO_2 -P25 suspensions in an aqueous solution of NaHCO_3 0.01 M and pbn (0.1 M) at pH 7.2–7.5.

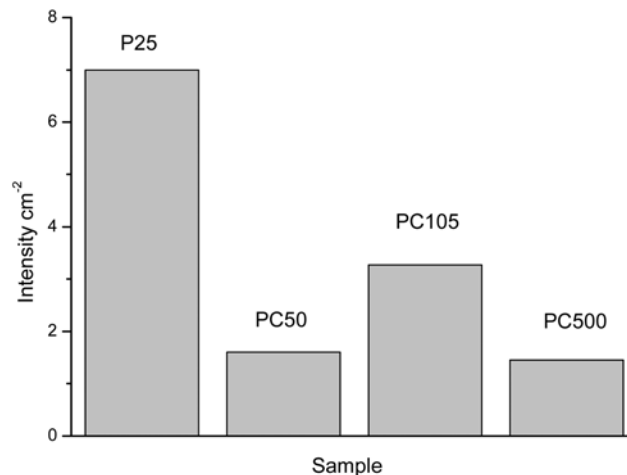
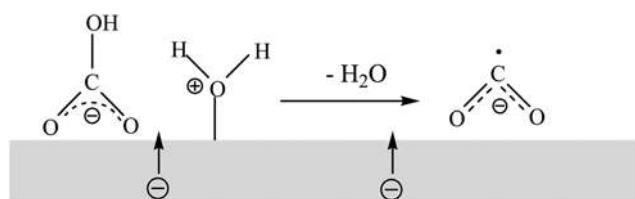


Fig. 6 Intensity of the $[\text{pbn-CO}_2]^\bullet$ adduct normalized by the area of the sample obtained on illumination (3 min) of different commercial TiO_2 powders dispersed in deaerated aqueous solutions containing NaHCO_3 (0.01 M) and pbn (0.1 M). pH 7.2–7.5.



Scheme 1 Proposed reaction scheme for reduction of the adsorbed bicarbonate in TiO_2 suspensions at pH 5–7.

In Fig. 6 we report the intensity of the $[\text{pbn-CO}_2]^\bullet$ adduct recorded at a fixed field position after 3 min illumination of some commercial TiO_2 samples in a HCO_3^- solution at a neutral pH. For a comparative analysis one must consider that the surface area is an important parameter and since it is significantly different for the examined samples (50 to $320 \text{ m}^2 \text{ g}^{-1}$), all signal intensities have been normalized to the known surface areas. It is the significantly higher amount of $\bullet\text{CO}_2^-$ radicals produced on P25 that meets the eye and results are seemingly not directly correlated with areas.

An analogous comparative analysis of $\text{OH}^\bullet$ production measured by coumarin hydroxylation, on different TiO_2 samples, showed that the activity of PC 500 is about one half that of PC50 despite the area of the former is 6 times higher. This is less surprising than might seem at first and, in fact, concerning surface reactions, it has been previously remarked that there can be real or apparent effects depending on the factors at the origin of the effect itself. If the surface structure and chemical composition (electronic factors) prevail, catalysis is “real”. Alternatively, if the origin of the effect is simply related to the surface area one can speak of “apparent” catalysis, since it vanishes as surface area effects are accounted for.³³

The lack of correlation between the activity and the surface area for reduction and oxidation processes calls for an explanation that relies on the contribution of different surface characteristics including defects and the surface density of OH groups.^{22,34}

In relation to the P25 data in Fig. 6, we limit ourselves to remarking that according to the literature,^{35,36} a likely explanation is the anatase/rutile mixed structure of this catalyst that helps charge separation thus reducing recombination. It has been previously remarked that P25 often turns out to be a more active catalyst than others having a higher specific surface area,³⁷ stressing the importance of good charge carrier separation.

Apart from P25, the other three samples examined are made of pure anatase and arguably the higher activity of TiO_2 -PC105 has still another explanation. We think it is presumably connected with an enhanced adsorption of bicarbonate anions on a comparatively high number of Lewis acid sites that characterizes this particular sample.³⁸

During the course of the above described experiments, we have never detected $\text{H}^\bullet$ radicals and neither were they observed in the previous literature,²⁰ but we cannot *a priori* exclude that $\text{HCO}_2^\bullet$ is the result of an indirect, fast reduction of HCO_3^- by $\text{H}^\bullet$, if reduction of water preferentially occurs instead of reaction (3). The pH dependent thermodynamic standard potentials involving species that are relevant to this work (HCO_3^- , H_2CO_3 , $\text{HCO}_2^\bullet$) are³⁹

$$E_0 = -0.078 - 0.0591 \text{ pH } (\text{HCO}_2^\bullet / \text{HCO}_3^-) \quad (11)$$

$$E_0 = -0.267 - 0.0295 \text{ pH } (\text{HCO}_2^\bullet / \text{H}_2\text{CO}_3) \quad (12)$$

$$E_0 = -0.156 - 0.0591 \text{ pH } (\text{HCO}_2\text{H} / \text{H}_2\text{CO}_3) \quad (13)$$

$$E_0 = 0.0 - 0.0591 \text{ pH } (\text{H}_2 / \text{H}_2\text{O}) \quad (14)$$

At pH 7, potentials for reactions (11)–(13) are comparable and more positive than E_{fb} (-0.57 V, vs. NHE), but slightly more negative than that of H_2O reduction, *i.e.*, -0.42 V (NHE), which is then expected to be favored. It is important to point out, however, that oxides do not take the reversible potential of the H_2 reaction because, unlike the case of metals, H_2 is not dissociatively adsorbed.⁴⁰ More generally, standard conditions do not quite represent real ones.

Metal-modified TiO_2

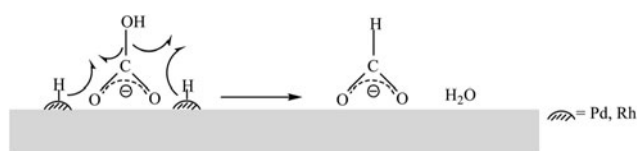
In the previous paragraph we concluded that we cannot prove or disprove reduction of HCO_3^- by $H^\bullet$. At least in short time-scale experiments, a mechanism involving these radicals may appear unconvincing due to the known kinetic limitations of H_2O reduction.

As mentioned above, H_2 does not dissociate on TiO_2 in neither the anatase nor the rutile structure modifications. The question of $H^\bullet$ radicals in TiO_2 is well described⁴¹ and it is claimed that TiO_2 can be used as an effective coating to slow down the dissociation rate of H_2 . However, once $H^\bullet$ is formed by reduction of water, it is mobile on the surface⁴¹ and its participation in the reduction of bicarbonate cannot be excluded *a priori*.

When TiO_2 is modified on the surface with some noble metal nanoparticles, the adsorbed hydrogen atoms ($H^\bullet$) are more stable and their role is clearer (Scheme 2).

In the case of $Pd(1\%)$ - TiO_2 the role of hydrogen in the reduction of HCO_3^- to $^{\bullet}CO_2^-$ is strikingly evident from the data of Fig. 7. The intensity of the $[pbn-CO_2^-]$ signal increases and reaches a plateau as a function of illumination time; on switching off the light at this point, one observes that the signal grows again. This effect can be interpreted as a manifestation of $H^\bullet$ participation in the reduction process that, as proposed earlier,¹⁵ is promoted by the ability of palladium to store hydrogen by forming stable metal-hydride bonds (Scheme 2). This property is in fact the basis of an interesting method of storage and generation of hydrogen by the bicarbonate/formate reaction.⁴² Analogous experiments with $Rh(1\%)$ - TiO_2 showed a comparable activity of $^{\bullet}CO_2^-$ formation under illumination, but the dark growth seen in Fig. 7 was absent. Both Pd and Rh can work as a sink of electrons but the important difference lies in the reported ability of the former to form more stable hydrides.

The last composite catalyst we examined was $Ru(1\%)$ - TiO_2 , which shows still another behavior. The EPR spectrum is



Scheme 2 Reduction of bicarbonate by hydrogen atoms formed in the discharge of $H_2O/H^\bullet$.

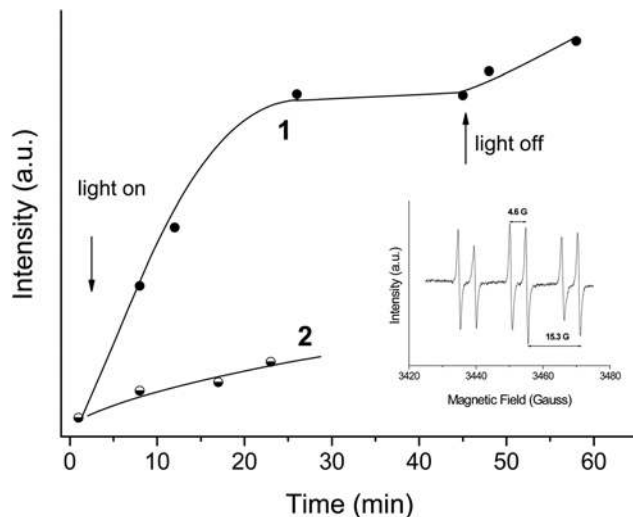


Fig. 7 Intensity of the $[pbn-CO_2^-]$ as a function of illumination time for $Pd(1\%)$ - TiO_2 (curve 1) and TiO_2 -P25 (curve 2) suspensions in an aqueous solution of $NaHCO_3$ 0.01 M and pbn (0.1 M) at pH 7.2–7.5. The inset shows the signal of the EPR signal at $t = 10$ min.

rather complex (ESI Fig. 4†) as it reveals not only trapped $OH^\bullet$ and $^{\bullet}CO_2^-$ radicals but also a $[pbn-CO_3^-]$ adduct (eqn (5)) whose signal intensity grows with illumination time even at pH 7. In this case, a likely explanation⁴³ is that initially Ru^0 sites are oxidized by holes to Ru^{n+} , *i.e.* hydrated species of the type $RuO(OH)_x$, which feature very good oxidizing power. These results demonstrate that the kind of metal chosen for surface modification cannot be arbitrary.

Experimental

Materials and methods

Several commercial TiO_2 samples were used in this study. Metal nanoparticle modifications of TiO_2 -P25 were obtained following the procedure described by Goren *et al.*¹⁵

Apparatus

EPR spin trapping experiments were performed with a Bruker ER 200MRD spectrometer equipped with a TE 201 resonator, at a microwave frequency of 9.4 GHz. Irradiation was carried out directly inside the EPR cavity with a Hanau Q-400 medium-pressure Hg lamp using a cut off filter ($\lambda > 360$ nm). The settings for the EPR spectrometer were as follows: center field, 3460 G; sweep width, 50 G; microwave frequency 9.4 GHz; modulation frequency 100 kHz; power, 6.3 mW; sweep time 41.94 s. The same flat quartz EPR cell was used for all measurements in order to minimize experimental errors.

Fluorescence spectra were recorded with a Jobin Yvon Spex Fluoromax II spectrofluorimeter equipped with a Hamamatsu R3896 photomultiplier. Both emission and excitation slits

were set at 5.0 nm during the measurements. Irradiation that precedes fluorescence measurements was performed with a Hg medium pressure lamp analogous to that described above. All the experiments were carried out in O₂-free suspensions at atmospheric pressure and at room temperature.

EPR spin trapping experiments

Concentrated suspensions of the photocatalyst (20 g L⁻¹) in aqueous solutions containing α -phenyl *N*-tert-butyl nitron (pbn, 0.1 M) were prepared both in the absence and in the presence of NaHCO₃ (0.01 M). In this last case, the initial pH of the solutions was 8.8. The suspensions were first deaerated with argon and pH was lowered by bubbling CO₂ for a time that depended on the requested pH value (4–7.5), as established by preliminary tests. Subsequently, the deoxygenated suspensions in a closed vessel were quickly transferred inside the flat EPR quartz cell inside a dry-box under an inert atmosphere. In the experimental pH interval 4–7.5, HCO₃⁻ is in equilibrium with dissolved CO₂ and we have checked that CO₂ loss into the gas phase during the transfer is negligible on the basis of pH measurements as well as using a CO₂ gas sensor (Thermo Scientific).

EPR spin trapping spectra were recorded as a function of irradiation time. No signals were obtained in the dark before irradiation. The amount of the employed spin trap is in large excess and it has been optimized on the basis of the dependence of the signal intensity on the spin trap concentration.

Fluorescence measurements

The samples were constituted by suspensions of the photocatalyst (about 3.5 mg mL⁻¹) in aqueous solutions containing coumarin (1.15 × 10⁻⁴ M) in the presence and in the absence of NaHCO₃.

A suspension (3 mL), at the desired pH value and deaerated as described above, was irradiated under magnetic stirring for 15 minutes ($\lambda > 360$ nm) in a Pyrex reactor (total volume 10 mL) provided with a 1 cm wide flat window. After irradiation the suspension was centrifuged and the fluorescence spectrum ($\lambda_{\text{exc}} = 332$ nm) of 7-hydroxycoumarin, eventually formed, was recorded ($\lambda_{\text{emiss}} = 455$ nm) using a standard fluorimetric quartz cell.

In order to verify the effect of possible pH changes during irradiation, parallel control experiments were carried out in the presence of diluted Na₂HPO₄–NaH₂PO₄ buffer. No difference was noted indicating that pH changes are negligible and that possible phosphate adsorption on TiO₂ does not affect formation of 7-hydroxycoumarin.

Conclusions

An EPR spin trapping investigation shows that parallel reactions take place in illuminated suspensions of TiO₂ in aqueous carbonate solutions (pH ≈ 7) which involve the formation of OH[•] radicals from photo-oxidation of water and carbon-oxygen

radical species resulting from the redox reactions of bicarbonate (HCO₃⁻). The salient results are

- The efficiency of formation of OH[•] radicals is proportional to the concentration of bicarbonate and depends on pH, as confirmed by an independent method using coumarin hydroxylation. In a weakly alkaline medium, photo-oxidation of HCO₃⁻ to [•]CO₃⁻ becomes evident. This process interferes with the oxidation of water.

- The EPR measurements show the formation of [•]CO₂⁻ radicals in suspensions containing bicarbonate at pH 5.5–7.5, confirming its reduction to formate as the initial stage. These radicals are not seen at pH 4 where dissolved CO₂ is the main species indicating that HCO₃⁻ in solution is the reducible species.

- The use of samples of different commercial origin shows that the efficiency of [•]CO₂⁻ formation is not directly correlated with the surface area of the photocatalysts. Allowing for this important parameter reveals that an explanation of the results must rely on the contribution of other factors related to the surface structure that condition adsorption phenomena and recombination of photogenerated charges.

- The effect of surface modification with noble metals on the amount of [•]CO₂⁻ detected depends on the nature of the metal. The radical is formed with high efficiency on illuminated Pd–TiO₂ and Rh–TiO₂; in the first case, the intensity of the EPR signal continues to grow after light is switched off strongly suggesting participation of hydrogen in the reduction of bicarbonate. Among the metal modified catalysts, Ru–TiO₂ features the peculiarity of favouring HCO₃⁻ oxidation. This is likely due to formation of oxidised surface species of the type RuO(OH)_x which, according to published literature, feature a very good oxidizing power. The results demonstrate that the kind of metal chosen for surface modification cannot be arbitrary.

In general, identification of radical intermediates of oxidation and reduction reactions can provide useful mechanistic information that may be used in the development of photocatalytic systems for the reduction of CO₂ also stored in the form of carbonates.

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